

Hunter Kruger-Ilingworth CP3406 Mobile Application Planning Document

Chosen Scenario

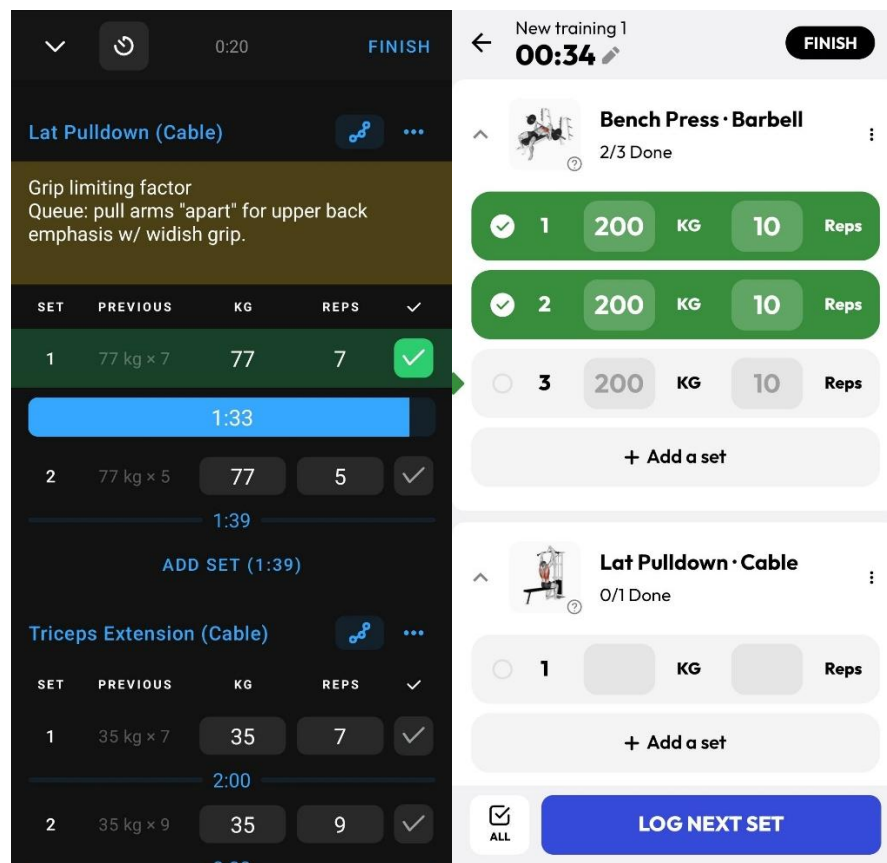
The intended user/scenario is an intermediate lifter who wants to track their gym progress. In strength training, applying progressive overload is a key indicator of strength. People who strength train have long turned to notebooks or apps to help track how strong they are in which exercises, but many existing apps are bloated with unnecessary features or hide basic tools behind subscriptions. As a user in this target audience, I find most current apps have features I never use, whilst features I would consider near essential are locked behind a subscription paywall.

I seek to create an app that trims the fat seen in current offerings – for a user who doesn’t want calorie or bodyweight tracking jammed into their workout app— as there are better tools for that. By focusing on core functionality, this app aims to do one thing well: track your strength progress – am I doing more weight than last week? Am I doing the same weight but increasing my reps? It is simple quantitative insights that answer questions like these that users of this app desire at a glance. Quality of life improvements are also required from offerings from the market as will be seen in the related apps section. By having one goal in mind, that is, strength tracking and nothing else, there is better chance I succeed in my goal given the application’s scope. Furthermore, the ‘min max’ philosophy of this vision would be better achieved if it can be used in tandem with other applications (through the use of android health connect API)

Related Apps

Below are the strengths and weaknesses identified for two mobile applications operating in the same market sector. Many strengths and weaknesses are shared between the apps analysed, but for brevity are not repeated. Currently, Strong is the application I’ve used

App Name	Strengths	Weaknesses	Improvements
“Strong” available on iOS and android	- Includes rest timers for structured sets - Clean and intuitive user interface (left figure below) - Supports custom exercises beyond the default list - Home screen provides a nice quick view on your workout consistency in the past few weeks - Calculates one-rep max estimates per set - Allows notes for both workouts and individual exercises - Effective rest timer integration - Data export functionality is available	- Poor tracking accuracy across different gyms; equipment inconsistencies (e.g. rusty cables) affect recorded performance, this can only be addressed in the workout by the user putting a note on the workout of what gym the session was performed at. - Historical performance is buried behind multiple UI navigation steps - Some gym machines lack unit markings (kg/lb), making it difficult to log load values accurately, and the app is unable to account for this - Body fat/body weight/measurement tracking is a feature that is 1. Locked behind a paywall, and 2. Is not of interest to my target user	- Add ‘unitless’ load measurement unit in addition to lbs and kg - Tag a workout with the gym you perform the workout to address equipment inconsistency - Make variations of an exercise be a subset of the core movement (some kind of hierarchical datatype)
“gym workout” app	- Very clean user interface (seen in rightmost figure below) - Like “strong” it contains tutorials on how to perform exercises, but is not a feature that would benefit the target audience of intermediate lifters I am to develop for	- Contains ads - No easy way to log variations of an exercise without creating a new custom entry - Variations on the same exercise e.g. Different grips are listed as completely different exercises leading to many repeated items when looking for your exercise leading to clutter in the ‘exercises’ page	Exclude body weight related features as it is outside the scope of a purely strength tracking application



Initial Feature List

User Features:

- Log in, persistent workout history attached to your account
- View and browse a history of past workouts stored locally
- Select from previously visited gyms or add a new gym when creating a workout session

Core Features:

- Create a new workout session with an automatic timestamp
- Add exercises to a workout, each containing a customisable list of sets
- Log each set with number of repetitions and load (in kilograms, pounds, or unitless for machines without weight markings)
- Automatically calculate training volume (weight × repetitions) and estimate [one-rep max](#) – display for each completed set
- View historical performance for a specific exercise, including ability to visualise trends over time whilst in the workout
- Integrate with [Android Health Connect](#) to allow third-party wellness apps to access logged workout data
- Potentially add [push notifications](#) for goal reminders (opt-in, as some users will find that annoying)

UI Elements:

- Use a numeric text input field composable for reps and load entry

- Use a drop-down element composable for weight unit selection (for an exercise), and gym selection (for a workout)
- Include an interactive rest timer with visual countdown (as seen in the leftmost figure above in blue)
- Display graphs and progress charts upon the user going into an exercise sub-menu, perhaps as an information icon

Learning Goals and Resources

Features/goals within scope:

- **Data visualisation** can be achieved through a library, some of which can be found on GitHub and the internet where you can make plots
 - Ycharts on jetpack compose <https://github.com/codeandtheory/YCharts>
 - Canvas API example From <<https://eevis.codes/blog/2023-07-24/more-accessible-graphs-with-jetpack-compose-part-1-adding-content/>>
 - Research will need to be done as to which is the best. Lots of other alternatives I found are essentially abandonware, and finding one that is up to date and can be used in Kotlin and jetpack compose may be a challenge
- **Android Health Connect** is a platform that enables apps to securely share and access health and fitness data on a user's device through a centralised system, this is used by the "Strong" app but not the "gym workout" application
 - This justifies removing extra features, as Health Connect lets users sync data to other fitness apps while using mine solely for strength logging.
 - Persistent storage of data could be handled through a [Room Database](#). Though if all data is stored in the Android health connect database this may not be necessary
 - Potential problems may arise in the specificity in the data I want to store (store the gym, store the units of an exercise etc)
 - Completing the [Learning modules / code labs](#) will help be out in resolving this confusion. The general dev page can be found [here](#)
 - If this ends up working, the user experience would be great, as it may have inherent cross-device data syncing if the integration with this API works as intended.

Features/Goals/Learnings out of scope:

Other potential features that, while nice, are probably out of scope for realistic implementation in this subject include

- Gym locations, opening hours provided by google maps API; [resources here](#); a regular user would have a regular routine and rarely check
- YouTube data API for YouTube video embedding (wouldn't add much); [resources here](#); pretty circumstantial use, may be useful if a user wants to embed a video about how to perform an exercise, but most people can just swap tabs
- Spotify API for music control within the app; [resources here](#); most people would prefer media control in the Spotify app itself